Bangladesh University of Business and Technology

Course Title: Computer Networks Lab
Course Code: CSE 320

LAB REPORT

Experiment Name: Wireless Routing

Experiment No.: **06**

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Introduction

Wireless Routing is the process of directing data packets between devices (nodes) over a wireless network to ensure efficient communication and high mobility. Common examples include Wi-Fi routing at home, Mobile Ad-hoc Networking (MANET), and Wireless Sensor Network (WSN) management. An example of home networking involves a wireless router connecting devices to the internet via 2.4 GHz, 5 GHz, or 60 GHz radio frequencies.

In this experiment, we will implement a basic wireless routing. We will further extend its functionality by connecting it to a polling router and adding a FTP server.

Devices and Tools

- Cisco Packet Tracer v9.0.0
- WRT300N Router × 2
- PC-PT × 6
- Laptop-PT × 1
- Smartphone-PT × 1
- Server-PT × 1

Procedure

The following steps were meticulously followed to setup static routing among the network and client devices:

Preparing the Network:

- In the Cisco Packet Tracer application, we add two new **Routers (WRT300N)**, eight client devices (six **PC-PT**, one **Laptop-PT**, one **Smartphone-PT**), and one **Server (Server-PT).**
- We interconnect **Router0** and **Router1** using the **Auto Cabling** tool.

Configuring the Wireless Routers:

- Single click **Router1** and a new window appears. Click the **GUI** tab to open the **Broadband Router Configuration Interface**.

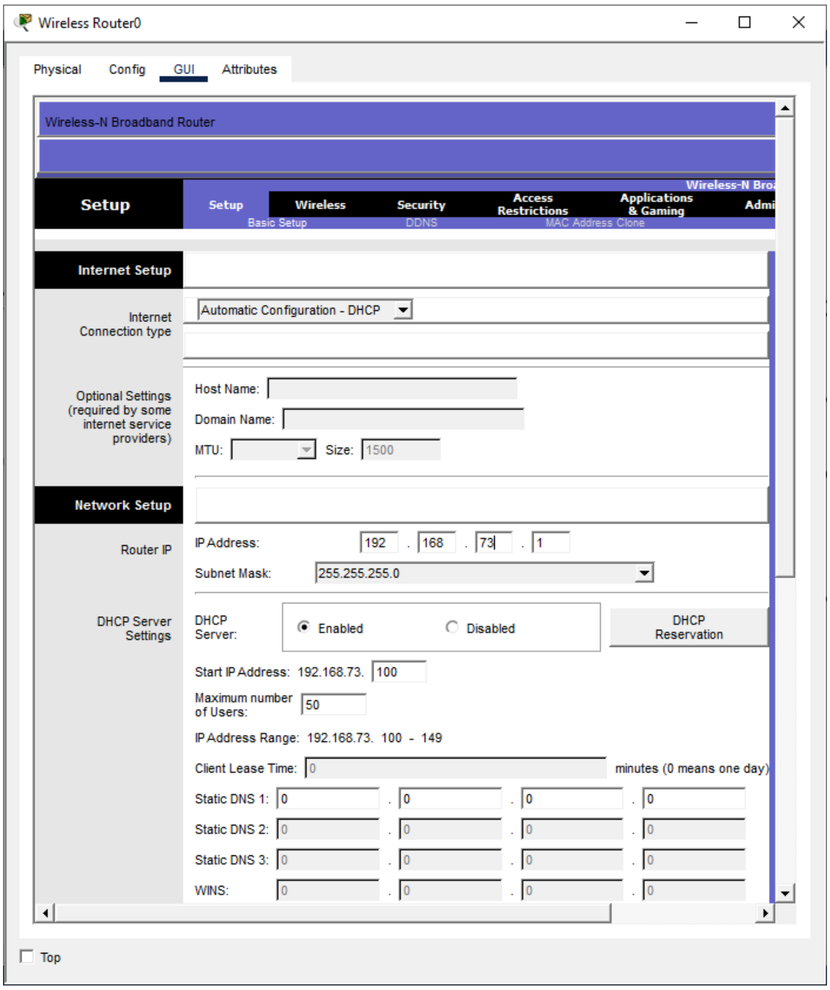

Fig 1.1: Broadband Router Configuration Interface.

- Under **Setup > Basic Setup** subtab, we configure the IP Address to **192.168.73.1**. We add the Subnet Mask to **255.255.255.0**. We also enable the **DHCP Server** by selecting the appropriate radio button. We leave everything else to their default settings.

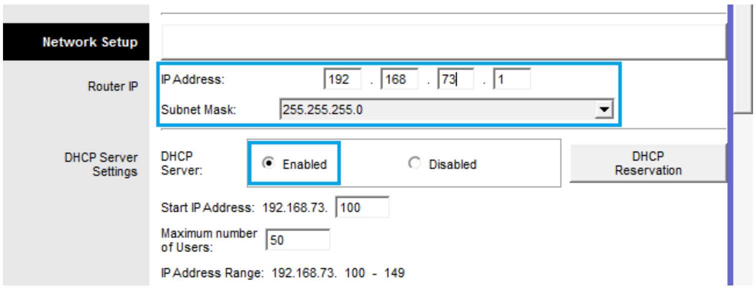

Fig 1.2: <u>Configuring the IP address and DHCP server for Router1.</u>

- Under **Wireless > Basic Wireless Settings** subtab, we set the **Network Name (SSID)** to **"Network A"**

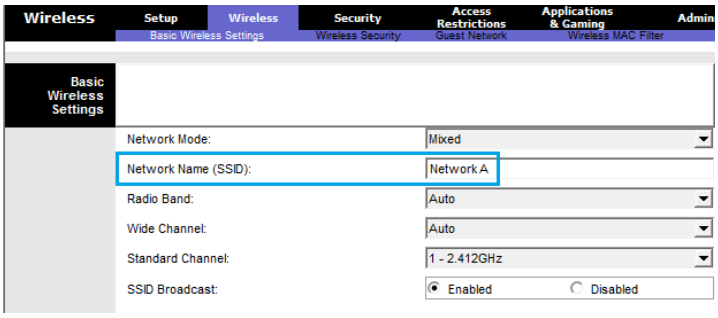

Fig 1.3: <u>Configuring the Network Name for Router1.</u>

- Under **Wireless > Wireless Security** subtab, we set the **Security Mode** to **"WPA2-Personal"** and **Passphrase** to **"BUBT1234"**

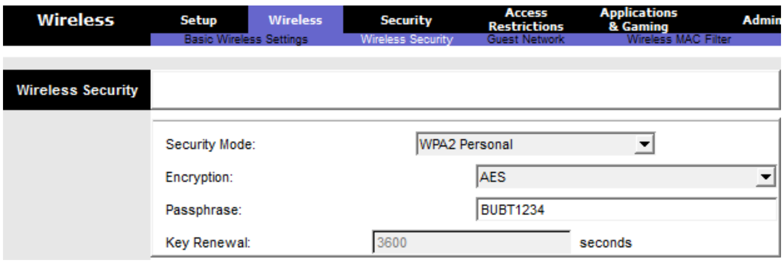

Fig 1.4: <u>Configuring the Security Mode and Passphrase for Router1.</u>

- Following the steps mentioned above, we configure **Router0** with a few changes. We set the **Network Name (SSID)** to **"Network B"** and we set the **Passphrase** to **"BUBT5678"**

- Since **Router0** and **Router1** are interconnected, they should belong under the same subnet to be able to share network traffic. To ensure that, we disable **DHCP Server** on **Router0**. In this way, **Router1** becomes the *polling* router. We also set the IP address of **Router0** to **192.168.73.2.**

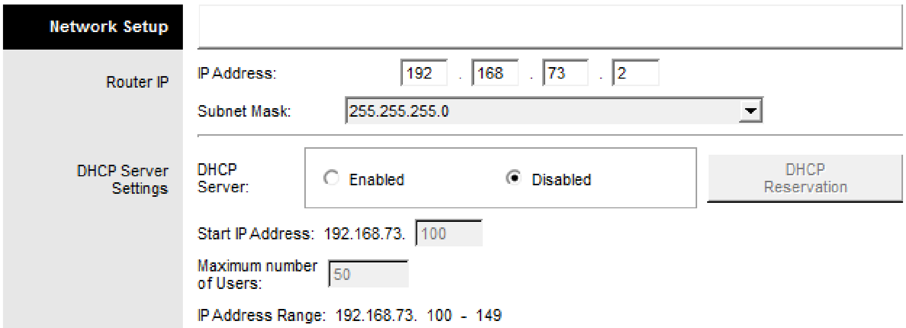

Fig 1.5: Setting up Router0 to belong under the same subnet.

- It is important to note that we must save the changes after every step. Otherwise, the changes will not take effect.

Fig 1.6: Saving the settings after each step.

Configuring the end devices:

- Single click the **PC0** and a new window appears.
- Under the **Desktop** tab, go to **PC Wireless**.

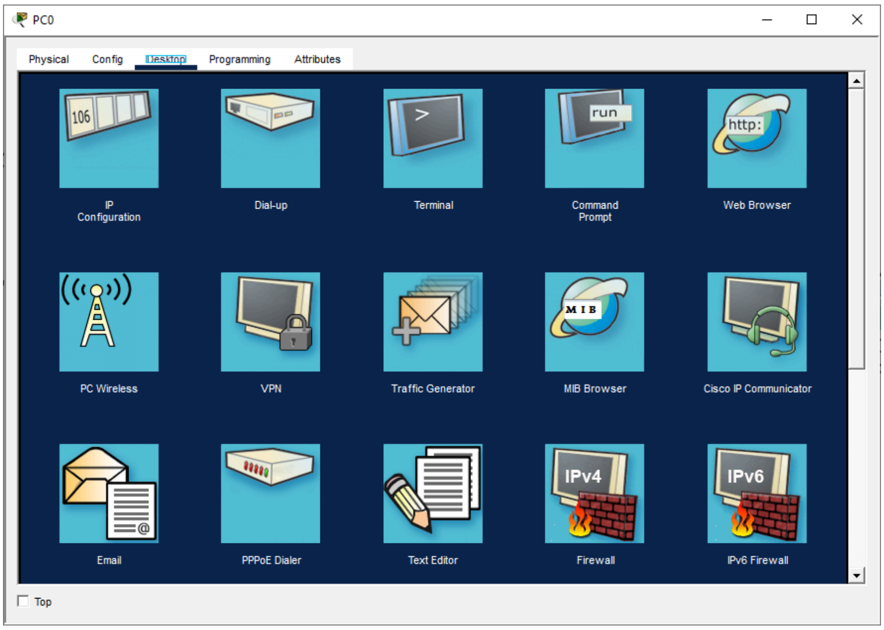

Fig 1.7: <u>Desktop interface for PC0.</u>

- A new window will appear, navigate to the **Connect** tab and press the **Refresh** button. We select **"Network B"** and press the **Connect** button. A new prompt appears. We enter the **Passphrase** of **Router0** in the field and press the **Connect** button again. If everything is okay, we should have successfully connected to **Router0**. We apply the same process to **PC1** and connect it to **Router0**.

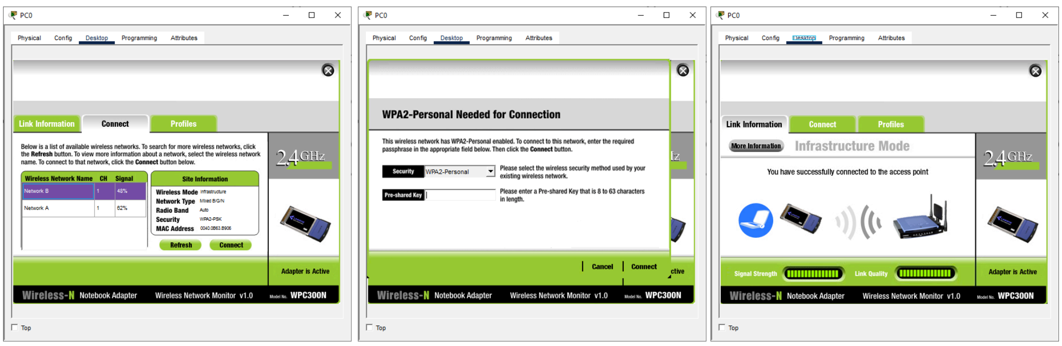

Fig 1.8: <u>Configuring the Wireless Connectivity of PC0.</u>

- Since we have connected **PC0** and **PC1** to **Router0**, we need to connect the remaining end devices to **Router1**. We follow the aforementioned steps but instead of selecting **"Network B"**, we select **"Network A"** and enter the **Passphrase** that was set in **Router1**. We follow the same steps to connect **Server0** to **Router1**.

- The Smartphone does not support a GUI interface where we can configure its Wi-Fi connectivity. So alternatively, we use the **Config** tab of **Smartphone0**. Under the **Interface** tab, we select **Wireless0**. We changed the **SSID** to **"Network A"**, Authentication to WPA2-PSK, and **PSK Pass Phrase** to **BUBT1234.**

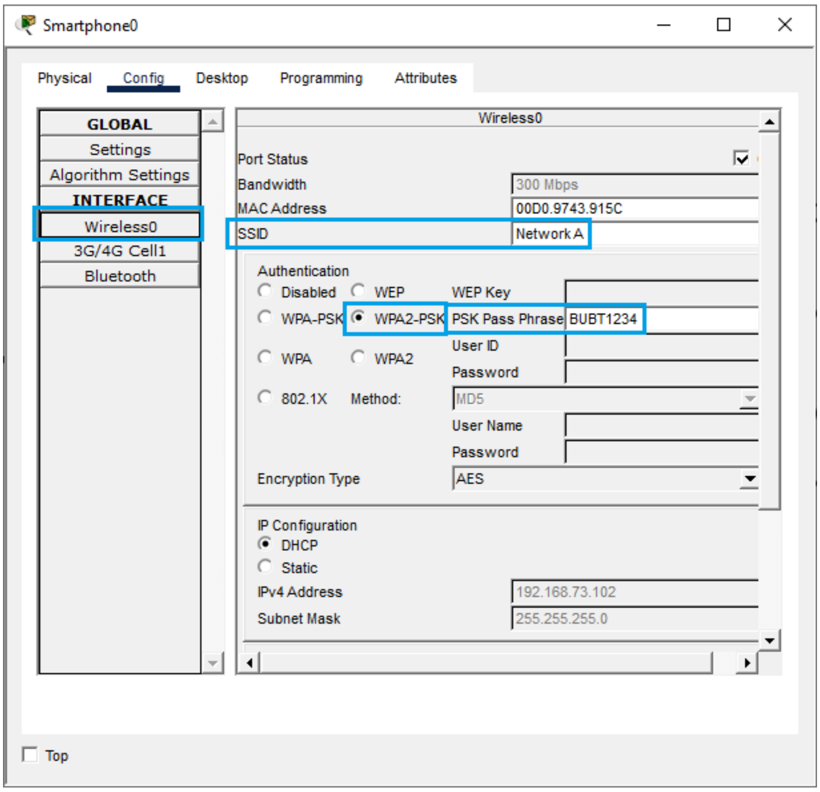

Fig 1.9: Configuring the Wireless Connectivity of Smartphone0.

Configuring the FTP Server:

- Click the **Server0**, a new window will appear.
- Go to the **Services** tab and select **FTP.** Turn on the **FTP Service**.
- Under the **User Setup**, type **bubt** in both Name and Address fields.

- Check **Write, Read, Delete, Rename,** and **List** permissions and click **Add**. This ensures that we would be able to login to the FTP server using the credentials we just added.

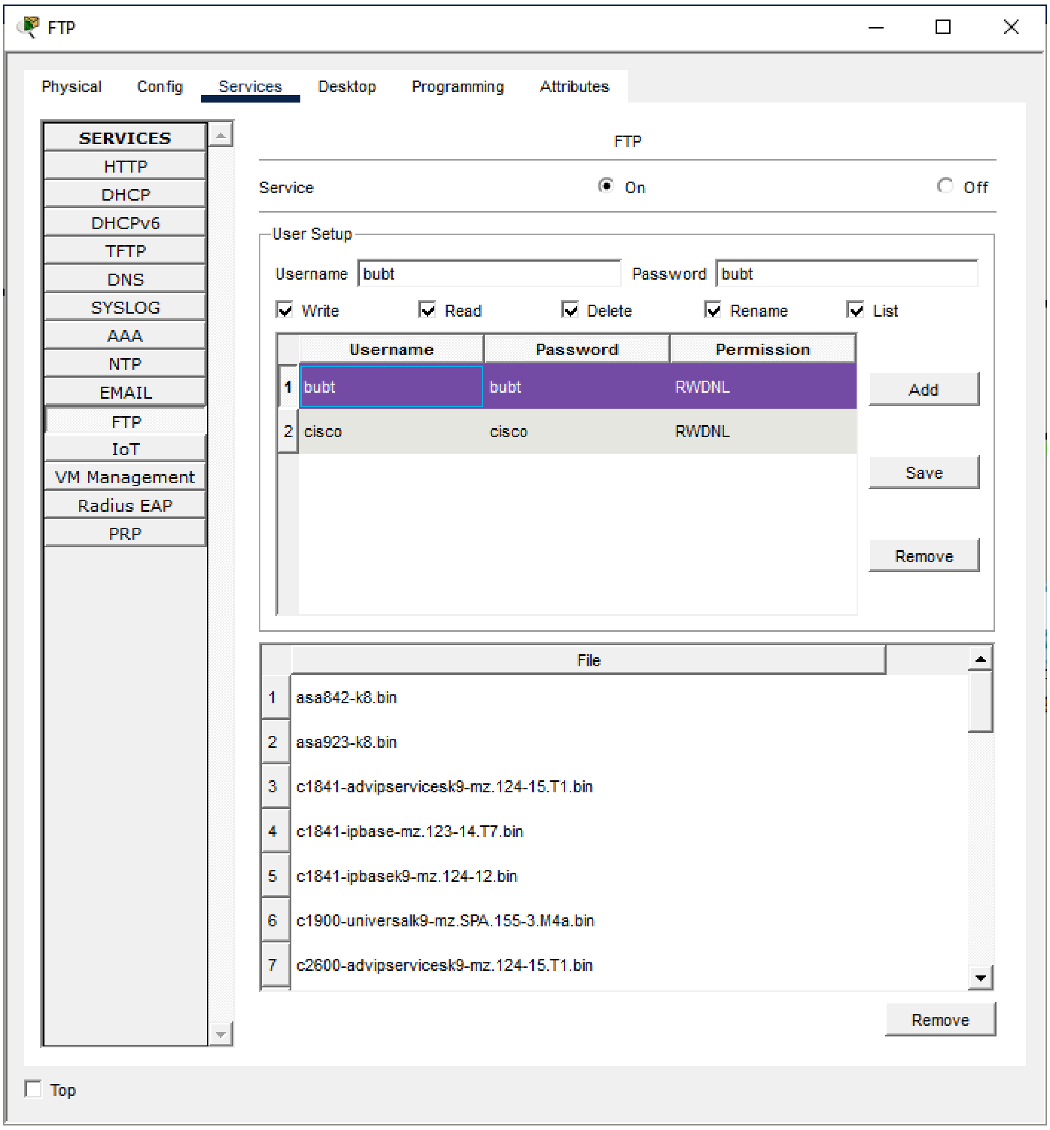

Fig 1.10: Configuring the FTP Server.

Uploading a file to the FTP Server:

- Click the **FTP server**, a new window will appear.
- Go to the **Desktop** tab and select **Text Editor**. Type **"BUBT"** in the text field. Hit **CTRL + S** to save the file.
- A new prompt will appear to enter the new file's name. Type **ONE_FILE.txt** and click **OK.**
- Close the **Text Editor** and select **Command Prompt**.

- Before logging in to the server, we look up its designated IP in the IP configuration.
- Login to the FTP server using the credentials.

```
C:\>ftp [IP of Server0]

Trying to connect… [IP of Server0]

Connected to [IP of Server0]

220- Welcome to PT Ftp server

Username: bubt

331- Username ok, need password

Password: ****

230- Logged in

(passive mod On)

ftp>
```

- Type the following command to upload the newly created file to the FTP Server:

```
ftp>put ONE_FILE.txt
```

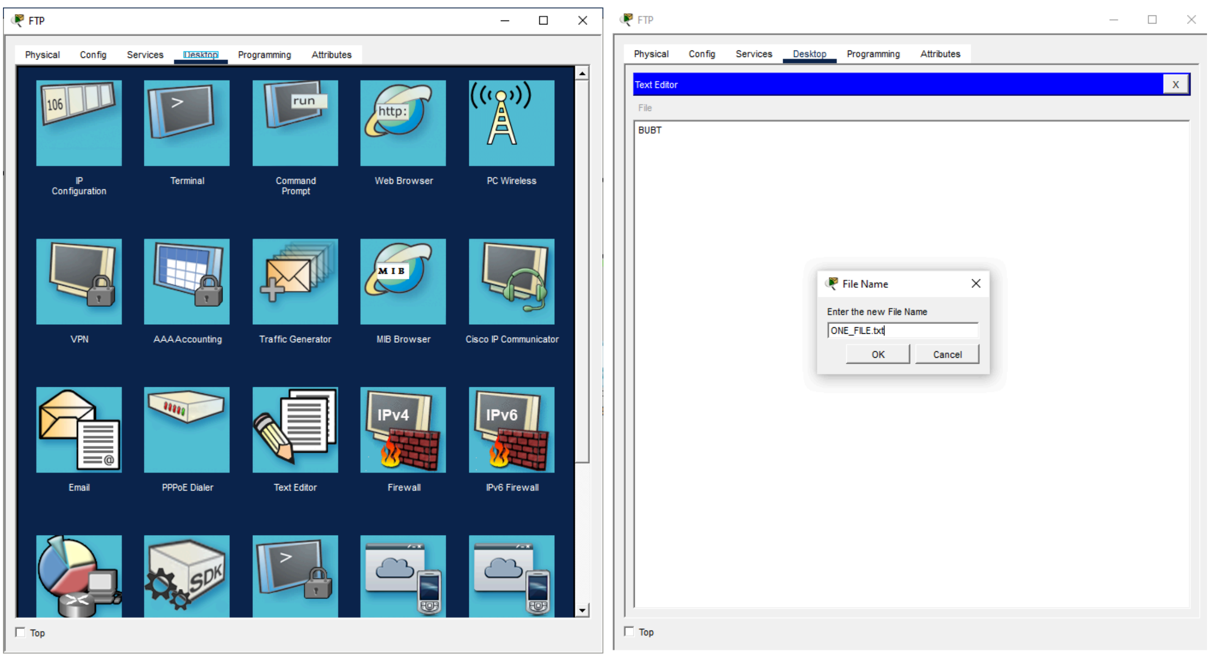

Fig 1.11: Creating a new text file.

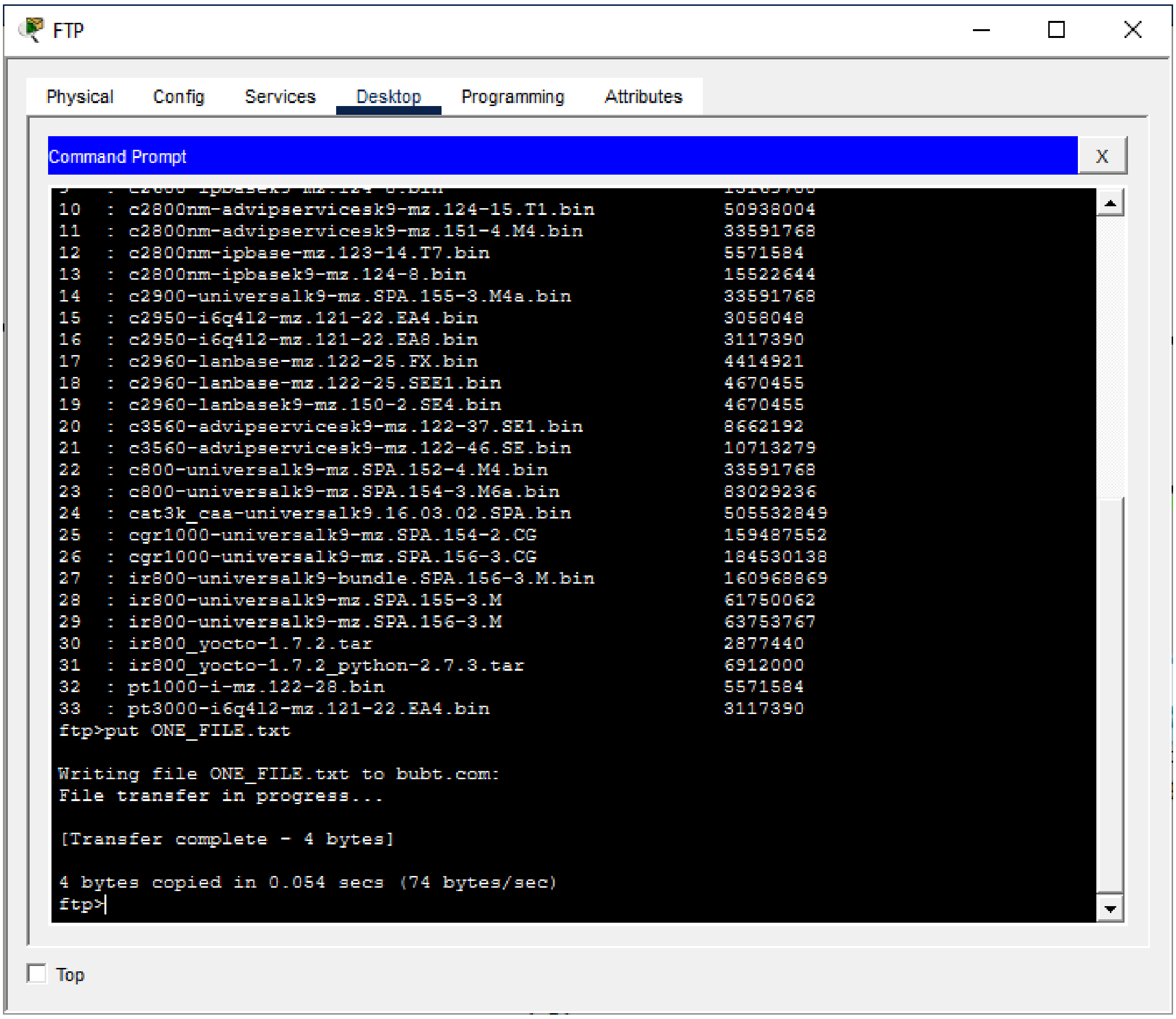

Fig 1.12: Uploading the text file to the FTP Server.

Downloading the file from the FTP Server:

- Click on any end device, a new window will appear.
- Under the **Desktop** tab, click **Command Prompt**.
- Type the following command to upload the newly created file to the FTP Server:

  `ftp>get ONE_FILE.txt`

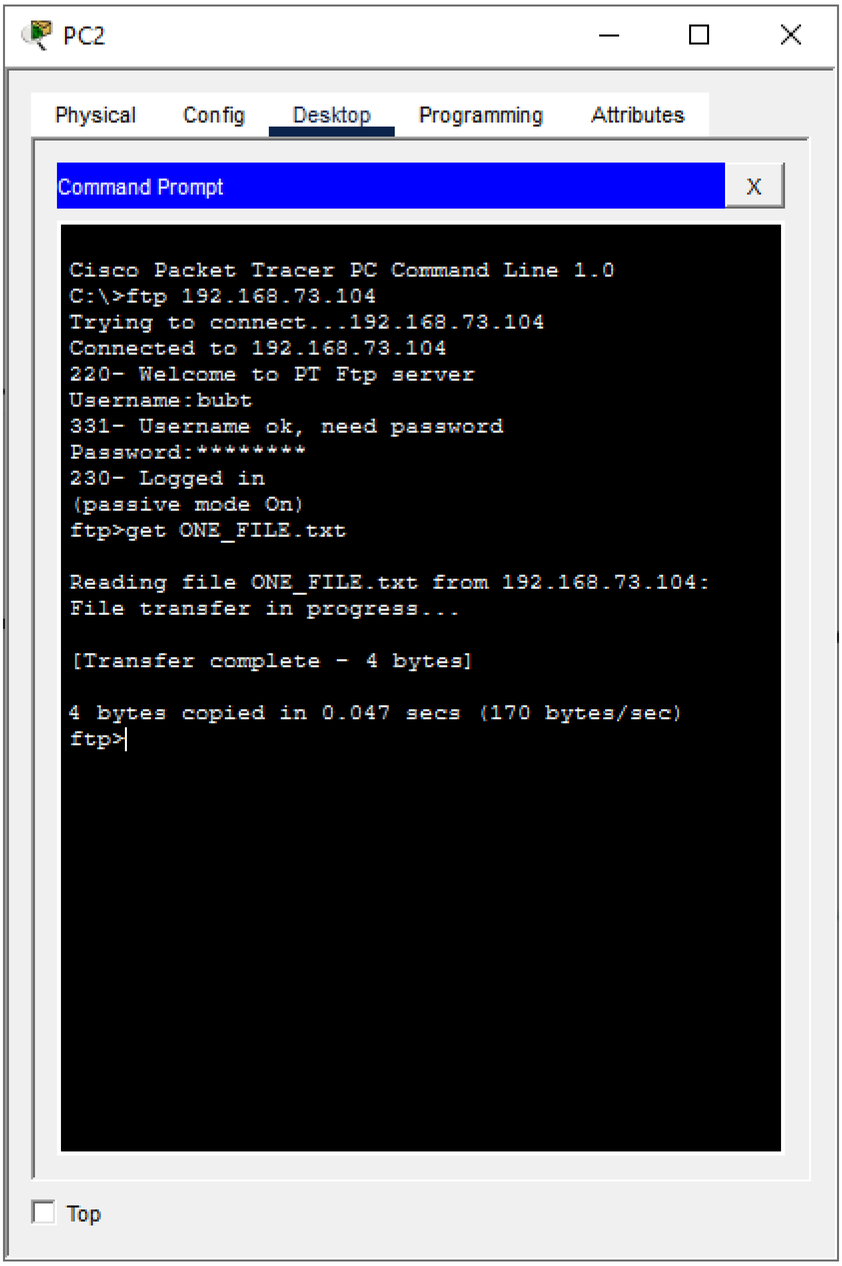

Fig 1.13: Downloading text file from the FTP Server to the client.

Result

Once we have followed all the steps mentioned in the previous sections, we should have a working Wireless Routing Configuration consisting of two routers, eight end devices and a FTP Server:

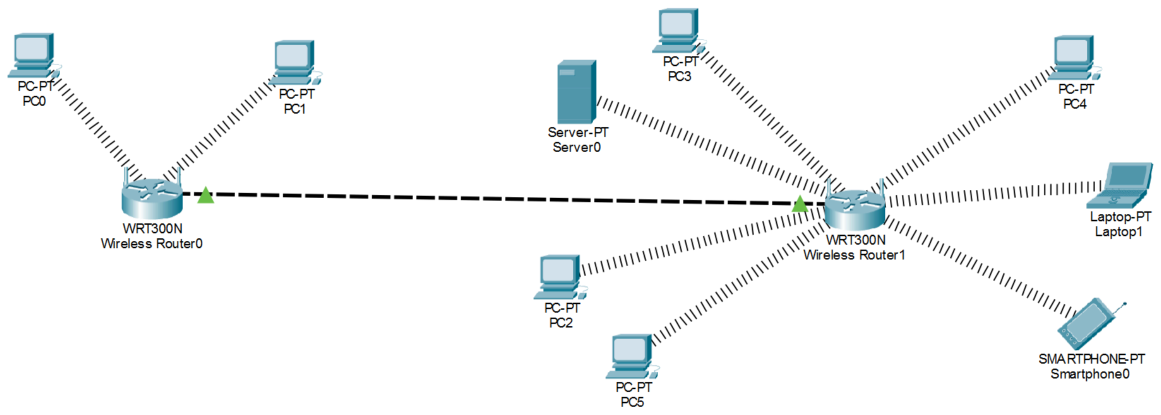

Fig 2.1: <u>Wireless Routing.</u>

Testing and Verification

We send messages from client to client, router to router, and client to server to check whether the data is being transmitted properly, ensuring that **Wireless Routing** is properly implemented.

- From the toolbar, click **"Add Simple PDU"**. The cursor would turn into a message icon.
- Click on **PC0** and then under the same router, click **PC1**.
- If everything goes as expected, we should see a successful communication.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Wireless Router0	Wireless Router1	ICMP		0.000	N	0	(edit)	
	Successful	PC0	PC1	ICMP		0.000	N	1	(edit)	
	Successful	Server0	PC3	ICMP		0.000	N	2	(edit)	
	Successful	PC3	Server0	ICMP		0.000	N	3	(edit)	

Fig 3.1: <u>Testing communication via Wireless routing.</u>

Conclusion

The objective of the sixth experiment was successfully achieved by configuring a Wireless Routing Configuration consisting of two routers, eight end devices and a FTP Server. It provided core skills for implementing wireless routing for home/office networks. We also learned the advantages of Wireless Routing and discovered a few caveats during the simulation. Since Cisco Packet Tracer has problems simulating connectivity between wireless routers, it is recommended to test the FTP server from a client PC that is under the same router.